

ELANCHEZHIAN P

Senior Technical Lead | Full Stack Engineer | .NET/C#, Next.js, NestJS, Azure, Multi-Tenant SaaS, AI/LLM

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Open to remote work across US, Canada, EU, and Australia time zones

PROFESSIONAL SUMMARY

Hands-on Senior Technical Lead with 6+ years of experience taking products from zero to one: architecting, building, and launching multi-tenant SaaS and enterprise products across .NET (C#, ASP.NET Core 8), modern JavaScript (Next.js, NestJS, React, Angular), Azure/AWS, and AI/LLM technologies. Provided technical leadership across a 20+ engineer cross-functional team delivering 6 concurrent products for clients in the United States, Canada, the Netherlands (Amsterdam), Australia, and the Middle East. Architected a HIPAA-aligned telemedicine platform handling 1,000+ monthly consultations, working with healthcare interoperability standards (HL7 FHIR, DICOM, SNOMED CT, ICD-10/11), and shipped production RAG/LLM features with human-in-the-loop design and cost controls. Practices AI-accelerated development (GitHub Copilot, Claude, Cursor) with human-reviewed quality gates, and secured 2nd place in the IronPDF Developer Challenge.

SELECTED CAREER HIGHLIGHTS

- Provided technical leadership across a **20+ engineer** cross-functional delivery team, mentoring 8 developers and coordinating technical delivery across **6 concurrent products** at a **95% sprint delivery rate**.
- Improved high-traffic API response times **30% at 50K+ daily requests** through Redis caching and async event-driven pipelines.
- Cut deployment cycles **50%** and production defects **40%** by instituting code-quality gates (SonarCloud, Sentry, Code Climate) and CI/CD.
- Reduced Azure cloud spend by **up to 40%** through cost-management analysis: consolidating databases into SQL Elastic Pools, scheduling VM auto-shutdowns, and automating restarts via Azure Workbooks.
- Modernized reporting workflows: cut report preparation from **2 hours to under 15 minutes** (an 87% reduction) via a Power BI star-schema data mart, and eliminated **15+ hours/week** of manual effort through SSIS/SSRS ETL automation.
- Architected multi-tenant platforms at scale: **1,000+ monthly telehealth consultations** (TeleMEDix) and **10K+ monthly assessments** (Scantron) with per-tenant row-level data isolation.
- Shipped a production **RAG pipeline** (Azure OpenAI, embeddings, vector search) with confidence-threshold human-in-the-loop routing and a per-feature LLM cost model.

CORE EXPERTISE

.NET / ASP.NET Core / C# • Next.js / React / Angular / TypeScript • Azure & AWS Cloud Architecture • Multi-Tenant SaaS & Event-Driven APIs • AI/LLM (RAG & Azure OpenAI) • Technical Leadership & System Design

TECHNICAL SKILLS

Backend: C#, ASP.NET Core (.NET 8 LTS), ASP.NET MVC, EF Core, REST APIs (OpenAPI), GraphQL, Node.js, NestJS, Express.js, Clean Architecture, DDD, SOLID

Frontend: Next.js (SSR/SSG/ISR), React, Angular (SPA/PWA, CDK, i18n, RTL/BiDi), TypeScript, Material UI, Tailwind CSS, Ant Design, HTML5/CSS3

AI / LLM: Azure OpenAI, RAG pipelines, LLM embeddings, vector search, semantic retrieval, streaming responses, human-in-the-loop design, per-feature LLM cost instrumentation; AI-assisted development with GitHub Copilot, Claude, Cursor, and prompt engineering under human-reviewed quality gates

Azure: Azure SQL, SQL Elastic Pools, App Service, Container Apps, Azure Functions, Service Bus, Event Grid, Blob Storage, API Management, Azure AI Search, Entra External ID (B2C), Key Vault, Application Insights, Azure Cost Management, Azure Workbooks, Front Door / WAF

AWS: S3, Lambda, EC2, RDS (PostgreSQL), Cognito

Data & Reporting: SQL Server, PostgreSQL, MySQL, MongoDB (self-managed standalone) with Percona PMM monitoring, Oracle, Redis, Power BI (star-schema data marts), SSIS/SSRS ETL, Chart.js, Highcharts, D3.js, query optimization, multi-tenant row-level security

Identity, Security & Payments: OAuth2/OIDC, JWT, multi-tenant RBAC, MFA, AWS Cognito, Azure Entra B2C, Stripe, PayPal, webhook-driven subscription billing, idempotency patterns

Healthcare (hands-on): FHIR, HL7, DICOM, HIPAA-aligned security and access-control design, clinical multi-role RBAC, tenant-level patient data isolation, telemedicine and care coordination architecture

Healthcare (working knowledge): SMART on FHIR, SNOMED CT, LOINC, RxNorm, CDA, ICD-10 / ICD-11

i18n & RTL: English, Arabic, Urdu, French, Angular i18n, CDK Bidirectionality, RTL-first layout design, Unicode and locale-aware data handling

DevOps & Quality: Docker, Azure DevOps CI/CD, GitHub Actions, environment provisioning, production deployments, domain mapping (DNS, SSL/TLS certificates), SonarCloud, Sentry, Code Climate, Jest with React Testing Library, xUnit, Selenium, Puppeteer

WORK EXPERIENCE

Senior Software Developer & Team Lead | App Innovation Technologies Pvt Ltd, Coimbatore Apr 2021 – Present

- Serve as technical SME across **6 concurrent products** for clients in the **US, Canada, Amsterdam (Netherlands), Australia, and the Middle East**, owning architecture decisions and code reviews, mentoring 8 developers, and providing technical leadership across a **20+ engineer** cross-functional team at a **95% sprint delivery rate**.
- Deliver products **from zero to one**: own greenfield setup end-to-end (solution scaffolding, cloud environment provisioning, CI/CD pipelines) through production deployment, **custom domain mapping (DNS, SSL/TLS)**, and go-live for client platforms.
- Design and ship REST APIs in **ASP.NET Core (.NET 8)** and **NestJS/Node.js** with OpenAPI documentation and **row-level security** (EF Core, Azure SQL, PostgreSQL); integrate Azure services end-to-end (App Service, Blob Storage, API Management, Service Bus, Event Grid, Functions, Application Insights) for event-driven, cost-instrumented workloads.
- Architect and deliver **Next.js**, React, and Angular TypeScript SPAs with shared component design systems (Angular CDK, Material UI, Tailwind, AntD) and full internationalization (**English, Arabic, Urdu, French**) with right-to-left bidirectional layouts.
- Champion an **AI-accelerated development workflow** adopted across the **20+ engineer team**, using GitHub Copilot, Claude, and Cursor for prototyping, code generation, test scaffolding, and review support, while enforcing human-reviewed quality gates for all AI-generated code.
- Architect multi-tenant **RBAC and identity** supporting 5+ user roles (patient, provider, practitioner, org admin, super admin) via OAuth2/OIDC with AWS Cognito and Entra B2C, including **MFA for privileged clinical roles**.
- Ship **AI-powered product features**: a production RAG pipeline with LLM embeddings and semantic search, plus AI-driven player-performance prediction for a sports analytics platform.
- Engineer for scale: Redis caching for LLM output and high-traffic APIs delivered **30% faster responses at 50K+ daily requests**; async pipelines on Azure Functions and Service Bus handle large-batch processing.
- Cut Azure cloud costs by **up to 40%** via Azure Cost Management analysis: consolidating databases into **SQL Elastic Pools**, scheduling VM auto-shutdowns, and automating restarts through Azure Workbooks; operate self-managed standalone **MongoDB** monitored with **Percona (PMM)**.
- Lead requirements gathering and delivery alignment with global stakeholders across 4 simultaneous engagements; enforce quality gates (SonarCloud, Sentry, Code Climate), cutting deployment cycles **50%** and production defects **40%**.

Technical Associate | Encore Theme Technologies Pvt Ltd, Chennai Aug 2019 – Apr 2021

- Developed financial web applications with ASP.NET MVC, C#, SQL Server, and Oracle under strict data-accuracy and compliance standards.
- Designed and optimized stored procedures, SSIS/SSRS ETL pipelines, and multi-tenant schemas, improving reporting efficiency **25%** and eliminating **15+ hours** of weekly manual effort.
- Contributed to SDLC documentation, requirements gathering, and production deployments in Agile teams.

KEY PROJECTS

TeleMEDix: Enterprise Telemedicine & Care Coordination Platform

React (TS) · ASP.NET Core · Clean Architecture · DDD · Node.js · GraphQL · PostgreSQL · Redis · Azure · AWS · Entra B2C · Cognito

- Architected a multi-tenant healthcare platform on **Clean Architecture and domain-driven design (DDD)**, scaling to **1,000+ monthly consultations** with a five-role clinical RBAC model (Patient, Provider, Practitioner, Admin, Org Admin) in a single unified application with zero code duplication across roles.
- Implemented **HIPAA-aligned safeguards**: OAuth2/OIDC identity (Cognito for patients/providers, Entra B2C for enterprise tenants), MFA for privileged clinical roles, and per-tenant row-level patient data isolation.
- Cut database query load **35%** and average patient-data wait times **25%** by introducing Redis caching over GraphQL and REST data layers; managed hybrid Azure + AWS infrastructure (Blob storage, WebJobs and Hangfire background processing, RDS PostgreSQL, EC2 microservices).

Scantron: Multilingual Survey, Assessment & Data Capture Platform

.NET 8 • ASP.NET Core • Azure SQL • Service Bus • EF Core • Angular • i18n/RTL • Power BI

- Designed an **asynchronous processing architecture** on Azure Service Bus and Functions to decouple high-volume submission and scoring workloads from synchronous APIs, maintaining stable user-facing latency during peak processing while scaling to **10K+ assessments monthly** with Container Apps auto-scaling.
- Delivered English, Arabic, and Urdu as first-class languages with full RTL bidirectional layouts and Unicode-aware rendering across all question types (MCQ, Likert, open-text, matrix).
- Reduced report preparation time **87%** (from 2 hours to under 15 minutes) across client and admin dashboards by building a Power BI star-schema data mart.

AI Document Chatbot: RAG-Powered Enterprise Q&A

Azure OpenAI • LLMs • Embeddings • Vector Search • Node.js • REST APIs • Redis

- Designed a complete RAG pipeline (document chunking, embedding generation, vector similarity indexing, semantic retrieval, LLM response synthesis) integrated into production REST APIs with streaming responses for low perceived latency.
- Controlled LLM API spend at scale through Redis caching of repeated queries and a **per-feature AI cost model** built by instrumenting every LLM call via Application Insights; applied confidence-threshold routing between automated and human review.

Multi-Tenant SaaS Portal: Multi-Vendor Subscription and Billing Platform

React (TS) • NestJS • PostgreSQL • Redis • Stripe • PayPal • Docker • AWS

- Architected a five-role SaaS portal (Vendor, Customer, Reseller, Account Manager, Super Admin) on a **NestJS** backend, giving each role distinct feature gating, usage quotas, and billing plans managed through a single codebase.
- Eliminated duplicate-payment risk with idempotency keys across Stripe and PayPal billing, and automated the full webhook-driven subscription lifecycle: plan upgrades, downgrades, trial expiry, and invoice generation.
- Enabled **zero-downtime plan changes** via a Redis-backed feature-flag and entitlement layer enforcing per-tenant plans without application restarts; deployed on AWS with Docker, RDS PostgreSQL, and S3.

Client Analytics Dashboard: Centralized Marketing Intelligence Platform

Angular • React (TS) • ASP.NET Core • Azure SQL • Azure Blob • Cognito/Entra • Chart.js • Highcharts • D3

- Cut client report preparation **from hours to minutes** by consolidating **6+ ad platforms and CRM tools** into one multi-tenant dashboard with scheduled data-refresh pipelines, role-based auth, per-client row-level isolation, configurable KPIs, and white-label theming.

Ballers: Sports Analytics & AI Player Performance Scoring Engine

REST APIs • PostgreSQL • AI/ML • Semantic Search

- Delivered **real-time, live-game stat aggregation** (shot attempts, makes, derived accuracy metrics per player/team) through performance-tuned aggregation pipelines over a multi-tenant schema, and integrated AI/ML services predicting player skill ratings from shot patterns.

FinFlow: Automated Financial Reporting & Compliance Platform

ASP.NET MVC • C# • SQL Server • Oracle • SSIS/SSRS • Stored Procedures

- Improved end-of-day reporting throughput **25%** and saved **15+ hours weekly** by tuning complex stored procedures, optimizing SSIS/SSRS ETL pipelines, and replacing 3 manual workflows with scheduled pipelines, all under strict accuracy, audit-trail, and regulatory compliance requirements.

ACHIEVEMENTS & RECOGNITION

- **IronPDF Developer Challenge, 2nd Place:** secured runner-up position in the global developer competition conducted by IronPDF.
- **NuGet package author:** published and maintain **HeicJpegToolkit** (v1.5.5, .NET 8, MIT-licensed), an open-source image-conversion library transforming HEIC files to JPEG, PNG, TIFF, GIF, WMP, and HEIF, with **1.4K+ downloads** on nuget.org (nuget.org/packages/HeicJpegToolkit).
- **Open source contributor:** merged bug fixes in NetiMobileDevice and HTML2Canvas (mainline releases).
- **Technical writing:** authored API design standards, developer onboarding guides, and sprint ceremony documentation adopted across multiple product lines.

EDUCATION

Bachelor of Engineering (B.E.), Electrical & Electronics Engineering

Anna University, Chennai, India

2015 – 2019